

Package ‘tirlogit’

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Title Modelling choice data with and without ties in R

Version 1.0.0

Description

Provides tools for analyzing different types of responses in Discrete Choice Experiments using the generalized rank-ordered logit model, allowing for the presence of ties in the data.

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References Xi, Y., Meulders, M., & Vandebroek, M. (2024). A Unified Framework for Choice Experiments: Integrating Strict Preferences and Ties. Submitted.

R topics documented:

health	2
InformationCriteria	3
LRtest	3
mode_long	4
mode_wide	6
recode	6
summary.tirlogit	7
tirlogit	7
wtp	9
Index	11

health

*Synthetic stated preference data for healthcare interventions***Description**

A generated dataset `health` containing simulated choices and priority rankings for medical interventions with different attribute, allowing for ties.

The synthetic data includes three alternatives (A, B, and C), each representing a distinct medical intervention. The profile of each intervention is a combination of seven attributes: the type, the success probability (i.e. effectiveness), and the frequency of adverse effects, of the intervention; severity of illness; patient responsibility for causing disease; time span until symptoms appear; age at onset of illness.

We simulate a scenario with 100 individuals, each receiving 5 choice tasks. For each task, a priority ranking is generated for the interventions, where a lower value in rank indicates a higher priority. For instance, rank 1 represents the highest priority, rank 2 a slightly lower priority, with larger values indicating progressively lower priorities. Additionally, choice data is generated to indicate which intervention(s) is/are preferred to be reimbursed, allowing for multiple best choices in cases where respondents have no strict preference.

Usage

```
data("health", package = "tirlogit")
```

Format

A data frame in wide format with the following columns:

id individual ID.

taskid choice task ID.

type.A, type.B, type.C type of the medical intervention for alternatives A, B, and C (1 = preventive, 2 = curative).

timeSpan.A, timeSpan.B, timeSpan.C time span until symptoms appear, (1 = after 20 years, 2 = after 5 years, 3 = within a year).

effectivenss.A, effectivenss.B, effectivenss.C the success probability (i.e. effectiveness) of the intervention (1 = 33% effective, 2 = 66% effective, 3 = 100% effective).

adverseEffects.A, adverseEffects.B, adverseEffects.C the frequency of adverse effects (1 = often, 2 = rarely, 3 = never).

severity.A, severity.B, severity.C severity of illness (1 = not severe, 2 = severe, 3 = lethal).

lifestyle.A, lifestyle.B, lifestyle.C responsibility of patient's lifestyle for causing disease, (1 = fully, 2 = partly, 3 = not at all).

ageGroup.A, ageGroup.B, ageGroup.C age at onset of illness (1 = aged 80-90, 2 = aged 60-70, 3 = aged 40-50, 4 = aged 20-30, 5 = aged 0-10).

rank.A, rank.B, rank.C reimbursement priority ranking for alternatives A, B, and C (lower values indicate higher priority).

best.A, best.B, best.C an indicator variable representing whether intervention(s) are considered for reimbursement (1 = yes, 2 = no).

InformationCriteria	<i>Information Criteria for the tirlogit model</i>
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Description

Calculates the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for an object of class `tirlogit`, based on the log-likelihood and the number of parameters.

Usage

```
## S3 method for class 'tirlogit'
AIC(object, ...)

## S3 method for class 'tirlogit'
BIC(object, ...)
```

Arguments

<code>object</code>	a fitted model of class <code>tirlogit</code> .
<code>...</code>	additional arguments (currently not used).

Value

For `AIC.tirlogit`, the AIC value. For `BIC.tirlogit`, the BIC value.

Examples

```
data("mode_wide", package = "tirlogit")
model_modeChoices<- tirlogit(multiBest ~ cost | dist + income , data = mode_wide,
                             intercept = FALSE, reflvel = "car",
                             ties = TRUE, type = "best", shape = "wide",
                             indexes = c("case", "choiceSetID"), alts.names = "alt",
                             threshold.dist = NULL, num.draws = NULL)

summary(model_modeChoices)
AIC(model_modeChoices)
BIC(model_modeChoices)
```

LRtest	<i>Likelihood ratio test for tirlogit models</i>
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Description

The `LRtest` function performs a likelihood ratio tests for two nested `tirlogit` models, to check whether one model provides a significantly better fit to the data than the other model.

Usage

```
LRtest(model1, model2)
```

Arguments

model1 a fitted model of class tirlogit.
 model2 a fitted model of class tirlogit.

Value

Summary of the likelihood ratio test, showing degrees of freedom, log-likelihood values, chi-square statistic, and p-value.

Examples

```
### Conduct a likelihood ratio test for nested models ###
data(mode_long, package = "tirlogit")
model1 <- tirlogit(rank ~ cost, data = mode_long, intercept = FALSE, reflevel = "car",
  type = "rank", ties = TRUE, shape = "long", alts.names = "alt",
  indexes = c("case", "choiceSetID"))
model2 <- tirlogit(rank ~ cost + freq, data = mode_long, intercept = FALSE, reflevel = "car",
  type = "rank", ties = TRUE, shape = "long", alts.names = "alt",
  indexes = c("case", "choiceSetID"))
LRtest(model1, model2)
```

mode_long

Synthetic stated preference data for transportation modes

Description

A generated dataset containing simulated choice data for transportation modes across multiple choice tasks. The dataset is structured in long format, where each row represents a specific transportation alternative (e.g., train, air, bus, and car) within a choice task for an individual. Each alternative is characterized by a set of attributes, including cost, in-vehicle and out-of-vehicle time, and frequency.

The data includes generated responses across six different types of choice tasks types, with responses in specific variables:

- **Best choice** (bestUnique): Indicate the most preferred alternative in each task, allowing only a single most preferred choice per task.
- **Best choice(s)** (best): Indicate the most preferred alternative in each task, allowing for multiple best choices.
- **Ranking not allowing for ties** (rankUnique): Provide a preference ranking for each alternative within a task, without allowing for ties.
- **Ranking allowing for ties** (rank): Provide a preference ranking that allows ties, where lower values indicate higher preference. Tied alternatives share the same rank.
- **Best-worst choice** (BestWorstUnique): Indicate one single most preferred and one single least preferred alternative in each task.
- **Best-worst choice(s)** (BestWorst): Indicate the most preferred and the least preferred alternative(s) in each task, allowing for ties.

Usage

```
data(mode_long, package = "tirlogit")
```

Format

A data frame in long format with the following columns:

case individual ID.

choiceSetID choice task ID.

alt name of the alternative (i.e. transportation mode, one of train, car, bus and air).

dist choice set specific variable, the distance of the trip.

cost alternative specific variable, monetary cost associated with the transportation mode.

ivt alternative specific variable, in-vehicle travel time.

ovt alternative specific variable, out-of-vehicle travel time.

freq alternative specific variable, frequency of availability for the transportation mode.

income choice set specific variable, household income.

urban choice set specific variable, a dummy variable indicating whether the trip involves a large city as either the origin or the destination (1 = yes, 0 = no).

bestUnique a dummy variable indicating whether the alternative is selected as the best alternative; only one alternative can be selected as the best in each choice task (1 = most preferred, 0 = otherwise).

best a dummy variable indicating whether the alternative is selected as the best alternative; multiple best choices can be given (1 = most preferred, 0 = otherwise).

rankUnique preference ranking of the alternative; each alternative is assigned a unique rank; rank 1 for the most preferred alternative.

rank preference ranking of the alternative; ties are allowed; rank 1 for the most preferred alternative(s).

BestWorstUnique an indicator variable representing whether the alternative is selected as the most preferred or least preferred alternative; only one best choice and one worst choice can be selected (1 = most preferred, -1 = least preferred, 0 = otherwise).

BestWorst an indicator variable representing whether the alternative is selected as the most preferred or least preferred alternative; multiple best and worst choices are allowed (1 = most preferred, -1 = least preferred, 0 = otherwise).

Source

This dataset is generated based on the ModeCanada dataset from the R package `mlogit`.

References

Croissant, Y. (2020). `mlogit`: Multinomial Logit Models. R package version 1.1-1. Available from: <https://CRAN.R-project.org/package=mlogit>.

mode_wide

*Synthetic stated preference data for transportation modes***Description**

A generated dataset containing simulated choice data for transportation modes across multiple choice tasks, in a wide format. This dataset is a wide-format version of [mode_long](#). Each row corresponds to a choice task. It includes choice set specific variables (`dist`, `income`, and `urban`), alternative specific attributes for each mode (`cost`, `freq`, `ivt`, and `ovt`), and simulated preference data across various scenarios in separate columns. For detailed descriptions of the dataset, see [mode_long](#). Additionally, this dataset also includes character string representations of each preference response.

Usage

```
data(mode_wide, package = "tirlogit")
```

Format

A data frame in wide format.

recode

*Encode categorical variables***Description**

The function is used to encode categorical variables in a data frame, based on the specified encoding type. It supports two encoding types: dummy coding and effect coding. The input data can be in either long or wide format.

Usage

```
recode(data, name, ctype, reflevel = NULL, ...)
```

Arguments

<code>data</code>	a data frame containing the variables to be encoded. The data frame can be in either long or wide format. Variables can be of type numeric or character.
<code>name</code>	a character vector of variable names to encode.
<code>ctype</code>	a character vector specifying the encoding type for each variable: "D" for dummy coding, and "E" for effect coding.
<code>reflevel</code>	a character vector specifying the reference levels for encoding. For variables of type character, the default <code>reflevel</code> is the alphabetically first level. For numeric variables, the default <code>reflevel</code> is the level with smallest numeric value.
<code>...</code>	additional arguments (currently not used).

Value

A data frame with the encoded variables based on the specified encoding types.

Examples

```
data(health, package = "tirlogit")
name <- c("type", "timeSpan", "adverseEffects")
ctype <- c("D", "E", "D")
reflevel <- c(1, 1, 2)
health_coded <- recode(data = health, name, ctype, reflevel) # data in wide format
head(health_coded)
```

summary.tirlogit	<i>Summary method for tirlogit model objects.</i>
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Description

Provides a summary of the estimation results of the fitted tirlogit model.

Usage

```
## S3 method for class 'tirlogit'
summary(object, ...)
```

Arguments

object	a fitted model of class tirlogit.
...	additional arguments (currently not used).

Value

Estimation results of the fitted tirlogit model, including:

- The model call, showing how the tirlogit model was specified.
- A table that includes the estimates of the coefficients, and the corresponding standard errors, z-values, p-values, and significance codes.
- The log-likelihood of the fitted model.
- McFadden R² (only relevant if Intercept = TRUE).
- Likelihood ratio test of the fitted model (vs. null model, only relevant if Intercept = TRUE).

tirlogit	<i>Analyze various types of discrete choice data with the Generalized Rank-Ordered Logit Model</i>
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Description

The tirlogit function is used to fit the generalized rank-ordered logit model to analyze various types of discrete choice data with and without ties, including the best choice(s), best-worst choice(s), full ranking data with and without ties, etc.

The function supports a three-part formula syntax similar to that of the mlogit package, which includes:

- alternative specific variables with generic coefficients
- choice set specific or individual specific variables with specific coefficients
- alternative specific variables with alternative specific coefficients.

Usage

```
tirlogit(
  formula,
  data,
  intercept = FALSE,
  refllevel = NULL,
  type = "rank",
  shape = "wide",
  alts.names = NULL,
  threshold.dist = NULL,
  indexes = NULL,
  num.draws = NULL,
  ...
)
```

Arguments

<code>formula</code>	a symbolic description of the model to be fitted.
<code>data</code>	an ordinary <code>data.frame</code> containing the variables in the model.
<code>intercept</code>	logical; whether the alternative-specific intercepts are included in the model.
<code>reflevel</code>	the reference level (the base alternative) for the utilities when estimating the model (i.e. the one for which the coefficients of choice set specific covariates are set to zero); The default <code>reflevel</code> is the alternative whose name appears first when sorted alphabetically from A to Z.
<code>type</code>	a character string specifying the type of the input choice data; "rank" for full (complete) rankings, "best" for best choice(s), "BW" for best-worst choice(s), "string" for general rankings in string format.
<code>shape</code>	a character string specifying the shape of the input data, either "long" or "wide". By default, <code>shape = "wide"</code> .
<code>alts.names</code>	only relevant if <code>shape = "long"</code> ; a character string specifying the name of the column that contains the names of the alternatives.
<code>threshold.dist</code>	a character string specifying the distribution for the threshold parameter. By default, <code>threshold.dist = NULL</code> , meaning the threshold is fixed, hence common for all individuals. If <code>threshold.dist = "lognorm"</code> , the threshold is assumed to be individual-specific, following a log-normal distribution.
<code>indexes</code>	a character vector specifying the names of the columns that contain the individual IDs and the choice task IDs. <code>indexes</code> can be omitted if both <code>shape = "wide"</code> and <code>threshold.dist = NULL</code> .
<code>num.draws</code>	only relevant if <code>threshold.dist</code> is not <code>NULL</code> ; numeric; the number of Halton draws for the maximum simulated likelihood estimation for the distribution of the individual-specific threshold.
<code>...</code>	additional arguments (currently not used).

Value

An object of class `tirlogit` containing model fit results including the estimated parameters, log-likelihood, the variance-covariance matrix of the estimated parameters, and the convergence status.

Examples

```
### Analyze the single best choice data ###
data("mode_long", package = "tirlogit") # Load the mode_long dataset
model_SingleBest <- tirlogit(
  bestUnique ~ cost | dist,      # Formula
  data = mode_long,             # Dataset used is mode_long
  intercept = FALSE,            # Intercepts are not included
  reflevel = "car",             # Reference level is set to "car"
  type = "best",                # Response type is "best"
  shape = "long",               # Data is in long format
  indexes = c("case", "choiceSetID"), # Specify individual ID (case) and choice set ID (choiceSetID)
  alts.names = "alt",           # Names of the alternatives are in column "alt"
  threshold.dist = NULL,        # A common threshold for all the individuals
  num.draws = NULL              # Not relevant since threshold.dist = NULL
)
summary(model_SingleBest)        # Summary of the fitted model

### Analyze the best choices with possible ties ###
data("mode_wide", package = "tirlogit") # Load the mode_wide dataset
model_Best <- tirlogit(
  best ~ cost | dist,           # Formula
  data = mode_wide,             # Dataset used is mode_wide
  intercept = TRUE,             # Intercepts are not included
  reflevel = "car",             # Reference level is set to "car"
  type = "best",                # Response type is "best"
  shape = "wide",               # Data is in wide format
  indexes = c("case", "choiceSetID"), # Specify individual ID (case) and choice set ID (choiceSetID)
  alts.names = "alt",           # Names of the alternatives are in column "alt"
  threshold.dist = NULL,        # A common threshold for all individuals
  num.draws = NULL              # Not relevant since threshold.dist = NULL
)
summary(model_Best)              # Summary of the fitted model
```

wtp

Compute Willingness-to-Pay (WTP) Estimates for a tirlogit Model

Description

Computes the willingness-to-pay (WTP) estimates for specified variables relative to a chosen cost variable in a fitted tirlogit model.

Usage

```
wtp(model, var.names, wrt, ...)
```

Arguments

<code>model</code>	a fitted model of class <code>tirlogit</code> .
<code>var.names</code>	a character vector specifying one or more variable names for which the WTP will be computed.
<code>wrt</code>	a single character string specifying the name of the cost variable (the variable with respect to which the WTP is calculated).
<code>...</code>	additional arguments (currently not used).

Value

A table summarizing the WTP estimates, including the estimate, standard error, t-statistic, p-value, and significance codes for each specified variable.

Examples

```
mode_long$time <- mode_long$ivt + mode_long$ovt
model0 <- tirlogit(formula = best ~ cost + freq + time, data = mode_long, intercept = FALSE,
  refllevel = NULL, type = "best", shape = "long",
  indexes = c("case", "choiceSetID"),
  alts.names = "alt")
summary(model0)
wtp(model0, var.names = "freq" , wrt = "cost") # Compute WTP for a single variable
wtp(model0, var.names = c("freq", "time") , wrt = "cost") # Compute WTP for multiple variables
```

Index

* datasets

health, [2](#)

mode_long, [4](#)

mode_wide, [6](#)

AIC.tirlogit (InformationCriteria), [3](#)

BIC.tirlogit (InformationCriteria), [3](#)

health, [2](#)

InformationCriteria, [3](#)

LRtest, [3](#)

mode_long, [4](#), [6](#)

mode_wide, [6](#)

recode, [6](#)

summary.tirlogit, [7](#)

tirlogit, [7](#)

wtp, [9](#)